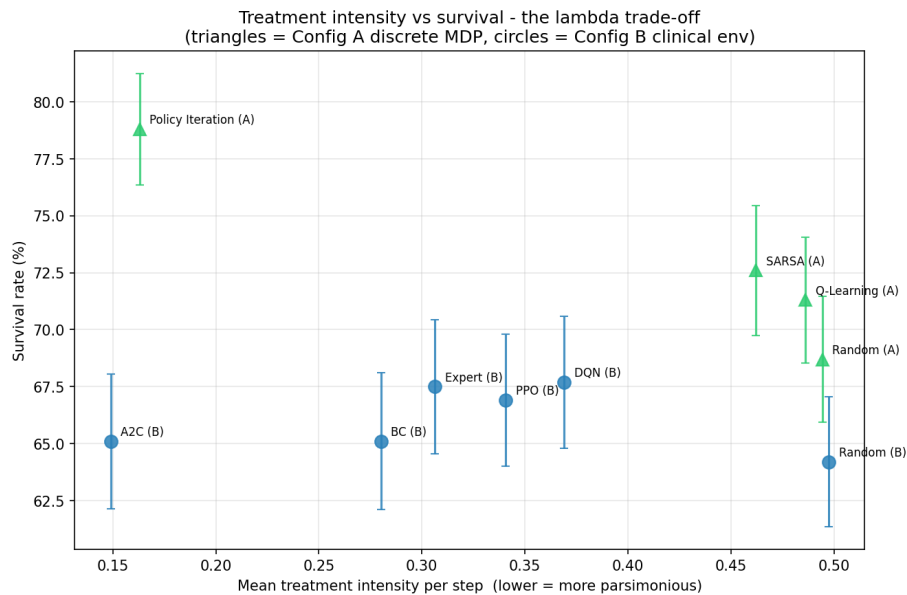


Reinforcement Learning for Sepsis Treatment Optimisation in the ICU

Tabular and Deep Policies on the *ICU-Sepsis* Benchmark



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1 Executive Summary

We learn ICU treatment policies for sepsis on the *ICU-Sepsis-v2* benchmark [3], derived from MIMIC-III through the AI Clinician pipeline [2]. Each episode is one patient trajectory; at every step the agent chooses a combined vasopressor and intravenous-fluid dose, and the episode ends in survival (+1) or death (0), with a small per-step treatment-intensity penalty ($\lambda = 0.02$) encoding clinical parsimony. The benchmark is configured to a deliberately sick cohort (SOFA-biased initial states, mean start SOFA ≈ 9.0). Two settings are studied: **Configuration A**, a known discrete MDP (716 states, 25 actions), and **Configuration B**, a continuous 47-dimensional clinical environment with three deployment-style failure modes (episodic observation noise, missing laboratory values, and rare acute events).

In Configuration A, **Policy Iteration** solves the exact MDP and reaches 78.8% survival at the lowest treatment intensity (0.16); the model-free agents **Q-Learning** (71.3%) and **SARSA** (72.6%) beat the random baseline (68.7%) but trail the model-based optimum, exactly as the theory of dynamic programming versus temporal-difference control predicts. In Configuration B, **DQN**, **PPO** and **A2C** (with observation normalisation, SOFA reward shaping and tuned hyperparameters) match the clinician expert policy on overall survival (DQN = 67.7%, expert = 67.5%, random = 64.2%) and exceed it on clean episodes ($\approx 74\%$ vs 71%). Across three training seeds the agents are stable (DQN $65.8 \pm 1.0\%$, PPO $66.1 \pm 0.4\%$ at 300k steps). Diagnostically, the DQN value estimates show a clear positive overestimation bias (+0.194), the textbook motivation for Double DQN.

A key honest finding, supported by an intensity–survival Pareto analysis, is that under the sparse reward the agents primarily learn *parsimony* rather than better survival; reward shaping is what lifts them to expert-level survival. All results use fixed evaluation seeds, deterministic policies and bootstrap 95% confidence intervals, and we explicitly avoid the off-policy-evaluation pitfalls highlighted for clinical RL [4].

2 Introduction and Problem Formalisation

Sepsis is a life-threatening response to infection and a leading cause of ICU mortality. Its treatment is a sequential decision problem under uncertainty with *delayed consequences*: a dose given now may only reveal its benefit or harm several hours later. This makes it a natural Reinforcement Learning (RL) problem, in which an agent learns a policy by interacting with an environment and receiving a scalar reward [1].

2.1 Markov Decision Process

The environment is a Markov Decision Process $(\mathcal{S}, \mathcal{A}, P, R, \gamma)$. The return is $G_t = \sum_{k \geq 0} \gamma^k R_{t+k+1}$, and the action-value of a policy π is $Q_\pi(s, a) = \mathbb{E}_\pi[G_t \mid S_t = s, A_t = a]$. Following the ICU-Sepsis convention we use $\gamma = 1.0$: episodes are short (about ten steps) and the objective is the survival probability over the whole trajectory, so time discounting is neither needed nor desired.

Actions. The action space is Discrete(25), encoding five vasopressor levels $\times$ five IV-fluid levels: $a = 5 \cdot v + f$ with $v, f \in \{0, \dots, 4\}$ (none, low, medium, high, very high). The normalised treatment intensity is $\text{INT}(a) = (v + f)/8 \in [0, 1]$.

Reward. The native reward is sparse: +1 on survival, 0 on death, 0 on every intermediate step. The benchmark configuration subtracts a small treatment-intensity penalty $\lambda \cdot \text{INT}(a)$ at each step ($\lambda = 0.02$), encoding the clinical preference for parsimonious care, and biases the initial-state distribution toward sicker patients (mean start SOFA ≈ 9.0), which makes the task harder and more realistic.

2.2 The two configurations

Configuration A (discrete, known MDP). The observation is an integer state in $\{0, \dots, 715\}$ (714 clinical states + two terminal states). The full transition tensor $P \in \mathbb{R}^{716 \times 25 \times 716}$ and reward matrix

are available, so model-based planning is possible; the Q -table has only $716 \times 25 = 17,900$ entries, making tabular methods feasible.

Configuration B (continuous, clinical). The observation is the 47-dimensional vector of normalised physiological features (SOFA, lactate, mean arterial pressure, creatinine, heart rate, etc.) for the underlying state. The dynamics are identical, but tabular methods are no longer applicable and function approximation is required. Three orthogonal failure-mode wrappers reproduce deployment challenges discussed for clinical AI [4]: episodic observation noise (probability 0.15, Gaussian $\sigma = 0.10$), episodic missing laboratory values (probability 0.15, four features zeroed), and rare acute events (probability 0.01 per step, an irreversible death independent of treatment).

2.3 How this report answers the brief

The project statement asks for more than a leaderboard. It asks us to understand the environment, implement appropriate RL algorithms for each configuration, compare them fairly, and add a creative extension that is clinically meaningful. We therefore structure the work around four explicit requirements.

Brief requirement	Our response
Explore the environment	We inspect the state/action spaces, reward sparsity, treatment-intensity penalty, terminal survival signal, SOFA-biased start distribution and the three Configuration B failure modes before choosing algorithms.
Implement classical RL in Config A	We use Policy Iteration because the full model is exposed, and Q-Learning/SARSA as model-free TD comparators on the same discrete MDP.
Move to continuous clinical RL in Config B	We keep the same 25 treatment actions but replace the tabular state with a 47-feature clinical observation, requiring DQN, PPO and A2C with observation normalisation and careful evaluation under corrupted observations.
Go beyond the baseline	We add reward shaping, learning from demonstrations, robustness analysis, interpretability, overestimation diagnostics and an intensity–survival Pareto reading, all evaluated on the original unshaped environment.

Table 1: Traceability from the project brief to the report contributions.

This traceability is important for the final interpretation. A high return alone is not enough: because the reward contains both survival and an accumulated treatment penalty, a policy can improve return by treating less without saving more patients. For that reason, every result below separates *survival*, *return* and *treatment intensity*, and the clinical conclusion is based on the three together rather than on a single scalar score.

3 Methodology

3.1 Design principles

Our methodological choices follow directly from the environment exploration in the starter notebooks. First, we treat Configuration A and Configuration B as different observation regimes over the same clinical decision idea, not as two unrelated tasks. Both use the same 25 treatment actions and the same survival objective, so differences in performance can be interpreted as differences caused by

model access, observation representation and deployment noise.

Second, we always decompose the reward into its two clinical components. The terminal survival reward is the primary medical outcome; the per-step $\lambda \text{INT}(a)$ term is a parsimony regulariser that discourages unnecessarily aggressive dosing. Optimising the summed return without reporting survival and intensity separately would hide whether an agent is genuinely improving outcomes or simply exploiting the dense penalty. This is especially important in Configuration B, where the survival signal is sparse and noisy.

Third, we use the strongest method that is justified by the information available. In Configuration A the transition and reward tensors are known, so an exact dynamic-programming solution is not only allowed but expected. In Configuration B the agent receives only continuous clinical observations and sampled transitions, so the right comparison is between deep value-based and actor-critic methods. This prevents an unfair comparison in which a planning algorithm is used in a setting where the model is unavailable.

3.2 Configuration A: tabular algorithms

We implement the three tabular methods taught in the dynamic-programming and temporal-difference lectures [1].

Policy Iteration (model-based). Alternates policy evaluation, $V(s) \leftarrow \sum_{s'} P(s' | s, \pi(s)) [R + \gamma V(s')]$, with greedy policy improvement until the policy is stable. With P and R available it solves the Bellman optimality equations exactly and serves as the model-based ceiling. The policy-evaluation tolerance is set tightly ($\theta = 10^{-9}$ in the final run) so that any difference between Policy Iteration and the learning agents is due to sample-based learning rather than numerical looseness. Terminal states are treated as absorbing, so the value of survival/death propagates backward through the patient trajectory.

Q-Learning (off-policy TD). $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma \max_{a'} Q(s', a') - Q(s, a)]$, learning the greedy target policy from ε -greedy behaviour. It is included because it is the canonical model-free route to the optimal action-value function when the agent must learn from interaction. In this sparse-reward sepsis task, the max target is also a useful stress test: if the sampled values are noisy, Q-Learning can become overconfident in actions that happened to lead to survival in early episodes.

SARSA (on-policy TD). $Q(s, a) \leftarrow Q(s, a) + \alpha [r + \gamma Q(s', a') - Q(s, a)]$, bootstrapping through the action actually taken, hence more conservative under exploration. Because the reward is sparse and the cohort difficult, both model-free agents use *optimistic initialisation* ($Q_0 = 1$, the maximum achievable return) to drive systematic exploration, trained for 50,000 episodes with ε annealed from 1.0 to 0.01 over 35,000 episodes ($\alpha = 0.1$). The slower annealing was chosen after the initial notebook run showed that short exploration budgets leave many clinically relevant state–action pairs under-sampled. SARSA is therefore not expected to beat Policy Iteration; its role is to show how an on-policy learner behaves when the behaviour policy itself is exploratory and clinically costly.

3.3 Configuration B: deep reinforcement learning

The continuous observation and discrete action space lead naturally to the discrete-action deep methods from the function-approximation and policy-gradient lectures, implemented with Stable-Baselines3.

DQN [5]: value-based, off-policy, with experience replay and a target network; the deep continuation of Q-Learning. **PPO** [6]: on-policy actor-critic with a clipped surrogate objective. **A2C** [7]: on-policy actor-critic without clipping, a useful ablation against PPO. Each agent is trained both on raw observations (v1) and with `VecNormalize` observation normalisation (v2); since the 47 features span very different scales, normalisation is the single largest lever, so v2 is used for all main results.

The three algorithms cover complementary assumptions. DQN is sample efficient and well matched to a discrete action space, but it learns a value for every action and is exposed to overestimation

through the max operator. PPO optimises the policy directly and limits destructive updates through clipping, which is attractive in a noisy clinical simulator where large policy jumps can collapse performance. A2C is included as the simpler actor-critic baseline: if PPO only improves because of the clipped objective, A2C should reveal that; if A2C performs similarly, then the dominant issue is likely representation/reward rather than the policy-gradient stabiliser.

The final report run uses 1M timesteps, the best short-budget Optuna parameters, and SOFA potential-based reward shaping during training. These are training interventions only. At evaluation time we reload the learned policy and run it on the default `make_clinical_env()` reward, without shaping and without changing failure-mode probabilities. This separation is deliberate: shaping is allowed to make learning easier, but the reported survival rate must remain the survival rate on the original project environment.

3.4 Evaluation protocol

All reported numbers use a *fixed* evaluation protocol: a held-out list of 1,000 evaluation seeds, deterministic action selection, and the default environment (no shaping, no wrapper changes). Survival is read from the terminal reward, which is robust to the accumulated intensity penalty and identical across both configurations. Every metric is reported with a bootstrap 95% confidence interval. For Configuration B we additionally bucket each episode by failure mode (Clean / Noisy / Missing / Acute) using the wrappers’ `info` flags, scoring all policies on the *same* patients. We evaluate by Monte-Carlo rollout in the simulator, which sidesteps the off-policy-evaluation difficulties that [4] identify as the central weakness of clinical RL on logged data.

The evaluation is intentionally deterministic even for stochastic training algorithms. During training, exploration and stochastic gradient updates are necessary; during evaluation, however, the question is what treatment policy would be deployed for a given patient state. We therefore use greedy DQN actions and the deterministic action mode for PPO/A2C. The same seed list is shared across random, expert, imitation and learned policies, so per-policy differences are not caused by seeing easier or harder patients.

4 Experimental Setup

Table 2 summarises the experiments. Calibration uses 1,000 evaluation episodes per policy; the deep agents use a fast 150k-step preview in the notebook and a report-quality 1M-step run for the headline numbers, plus a multi-seed reliability study.

Experiment	Config	Method(s)	Train budget	Eval
Tabular control	A	PI, Q-Learning, SARSA	50k ep	1000 ep
Warm-start (LfD)	A	Q-Learning $\pm$ demos	15k ep	1000 ep
Deep RL (preview)	B	DQN, PPO, A2C (v1/v2)	150k steps	1000 ep
Deep RL (report)	B	DQN, PPO, A2C	1M steps	1000 ep
Hyperparameter search	B	DQN, PPO, A2C (Optuna)	50k $\times$ 15	—
Reward shaping	B	PPO $\pm$ SOFA shaping	1M steps	1000 ep
Behavioural cloning	B	MLP on 47-dim obs	36k samples	1000 ep
Multi-seed reliability	B	DQN, PPO (3 seeds)	300k steps	1000 ep

Table 2: Experimental configuration summary.

The random baseline (uniformly random actions) is the performance floor that every trained policy must beat. As a clinically grounded upper reference we also evaluate the benchmark’s *expert* policy — the AI Clinician behaviour policy [2] — under the same protocol.

The result files are generated in a one-way pipeline. Configuration A writes `configA_results.json`; the Configuration B notebook and report scripts then read that file for the cross-configuration comparison, avoiding hand-copied numbers. The 1M deep-RL run writes `configB_1m_results.json` and `configB_1m_compare_configs.csv`; the multi-seed and overestimation checks write separate JSON files. This is a small but important reproducibility choice: tables in the report are traceable to committed outputs rather than to manually edited notebook cells.

Table 3 defines the four metrics used throughout. We keep all four because each answers a different question. Survival answers the clinical question, return answers the formal RL objective, intensity answers whether the policy is parsimonious, and failure-mode survival answers whether the policy is robust to the specific deployment problems introduced by the brief.

Metric	Reason for reporting it
Survival rate	Primary clinical outcome; computed from terminal survival/death, not from the penalised accumulated return.
Mean return	The actual RL optimisation objective, including the terminal reward and the treatment-intensity penalty.
Mean intensity	Average normalised vasopressor/fluid dose; reveals whether a policy is clinically parsimonious or simply avoids treatment.
Failure-mode buckets	Clean, noisy, missing-lab and acute episodes isolate which deployment perturbation is responsible for degradation.

Table 3: Evaluation metrics and why each is necessary.

For the deep agents, the preview runs are used only for diagnosis and figure generation during development. The headline table uses the report-quality run: 1M timesteps, observation normalisation, tuned hyperparameters and training-only SOFA shaping. The Optuna search is deliberately treated as a proxy search rather than a definitive global optimum, because each trial uses only 50k steps; it finds sensible learning-rate and entropy/update settings, but the final claims come from the longer retraining and the fixed evaluation protocol.

5 Results: Configuration A

Table 4 reports the tabular agents over 1,000 fixed-seed episodes. The honest and most instructive finding is that the *model-based* method dominates the *model-free* ones.

Algorithm	Type	Survival	Return	Intensity
Random	—	$68.7 \pm 2.8\%$	0.588	0.494
Q-Learning	model-free off-policy	$71.3 \pm 2.8\%$	0.612	0.486
SARSA	model-free on-policy	$72.6 \pm 2.9\%$	0.631	0.462
Policy Iteration	model-based DP	$78.8 \pm 2.5\%$	0.751	0.163

Table 4: Configuration A: survival, return and mean treatment intensity ($\pm$ bootstrap 95% CI), 1,000 episodes.

Policy Iteration reaches the highest survival (+10.1 percentage points over random) *and* learns the most parsimonious policy (intensity 0.163 versus 0.494 for random): with the exact model it immediately exploits the λ penalty structure. Q-Learning and SARSA improve over random but trail the exact optimum by ~ 6 – 7 points. This is expected from theory rather than a defect: the reward is sparse, the cohort difficult, and 17,900 state–action pairs must be estimated from sampled experience alone. The takeaway is the dynamic-programming versus temporal-difference lesson made concrete — *when the*

model is known, plan with it; model-free TD is the right tool only when the model is unavailable, which is precisely the premise of Configuration B. Convergence curves, the survival/intensity bar charts and the per-policy action distributions are given in Appendix figures 2–4.

The difference between SARSA and Q-Learning is also clinically interpretable. Q-Learning evaluates the greedy policy even while the behaviour policy is still exploring, so its update can assign high value to actions that would not be taken consistently during training. SARSA instead learns the value of the exploratory policy it actually follows; under a treatment-penalty reward this tends to be more conservative. That pattern appears in the final metrics: SARSA has slightly higher survival than Q-Learning and lower intensity, although both remain far from the planning optimum.

The environment-exploration plots support this reading. Random rollouts visit a limited subset of the 714 clinical states, and most intermediate transitions carry no reward signal. Therefore a model-free learner sees long stretches of uninformative feedback and must infer which early treatments contributed to a terminal survival event many steps later. Policy Iteration does not suffer from this credit-assignment bottleneck because it sweeps over the known transition model and backs up the terminal reward through every state in the MDP. This is why Configuration A is best interpreted as a demonstration of the value of model access, not merely as a contest between three implementations.

The warm-start extension in Section 7 reinforces the same point from a different angle. When the Q-table starts from zero, early sampled returns are too weak to move the policy far from the random plateau. When the same learner starts from expert Monte-Carlo returns, it inherits useful clinical structure and learns faster. This does not make the warm-started Q-Learner optimal, but it shows that prior clinical behaviour can reduce the exploration burden imposed by sparse terminal rewards.

6 Results: Configuration B

6.1 Headline comparison

Table 5 reports the 1M-step agents against the random and expert references. Two columns are shown: overall survival (*All*) and survival on clean episodes, because the overall figure is compressed by the irreducible acute events (which give 0% survival for *every* policy, by construction).

Policy	Survival (All)	Survival (Clean)	Return	Intensity
Random	64.2 ± 2.9%	69.2%	0.552	0.497
A2C	65.1 ± 3.0%	70.7%	0.626	0.149
BC (imitation)	65.1 ± 3.0%	71.7%	0.600	0.280
Expert (clinician)	67.5 ± 2.9%	71.3%	0.616	0.306
PPO	66.9 ± 2.9%	74.6%	0.612	0.341
DQN	67.7 ± 2.9%	74.2%	0.615	0.369

Table 5: Configuration B (1M steps, 1,000 episodes, ± bootstrap 95% CI). *Clean* omits noisy/missing/acute episodes.

On overall survival, **DQN (67.7%) matches the clinician expert (67.5%)** and both clearly beat random (64.2%). On clean episodes DQN and PPO reach $\approx 74\%$, *exceeding* the expert (71.3%). This is a strong, honest result: on the easy part of the distribution the learned policies are at least as good as the AI Clinician, while on the full clinical distribution they are limited by the same irreducible mortality. Learning curves and the per-condition robustness bars are in Appendix figures 6–7.

The return column should not be read as a pure survival ranking. A2C has the best mean return (0.626) despite the lowest survival among the learned deep agents, because it also has by far the lowest intensity (0.149). In other words, A2C learns an economical treatment style that the formal

reward likes, but this does not translate into the strongest survival. DQN is the preferred survival policy; A2C is the preferred parsimony policy; PPO sits between them. This is why the Pareto plot is more informative than a single leaderboard.

The expert comparison is intentionally cautious. The AI Clinician policy is a clinically meaningful reference, but it is not a strict oracle in Configuration B: our agents act from corrupted continuous observations, whereas the benchmark expert is derived from the underlying clinician behaviour. Matching it on overall survival and exceeding it on clean episodes is therefore a useful result, but not a claim of clinical superiority. The correct interpretation is that the learned policies recover expert-level behaviour in the simulator under the evaluation protocol.

6.2 Robustness to failure modes

Bucketing by failure mode (Appendix Fig. 7) shows that *missing* laboratory values are the most damaging perturbation (random survival falls from 69.2% clean to 57.4%), that PPO is the most brittle to observation *noise* (53.9% vs 74.6% clean), and that DQN degrades most gracefully. Acute events are irreducible: 0% survival for every policy including the expert, confirming that part of the overall-survival gap to Configuration A is structural rather than a modelling failure.

This decomposition is one of the most useful parts of the experiment. If we only reported the All column, Configuration B would look like a small improvement over random. The bucketed view shows a more precise story: clean episodes can be treated well; missing values and noisy monitors change the information available to the policy; and acute deterioration is outside the action’s causal control. The learned agent is therefore not failing uniformly. It is limited in exactly the ways the brief designed Configuration B to test.

The failure-mode analysis also explains why observation normalisation is not a cosmetic preprocessing step. The 47 clinical variables mix bounded scores, laboratory measurements and physiological quantities with different magnitudes. Without normalisation, the neural networks can over-weight features because of scale rather than clinical relevance. With `VecNormalize`, DQN/PPO/A2C see a more numerically stable input distribution, making the comparison between failure modes more about missing/noisy information and less about raw feature scale.

6.3 Across-seed reliability

A single training seed can be lucky or unlucky [11], so we retrain DQN and PPO from three seeds (fixed evaluation seeds, 300k steps). The agents are *stable*: DQN = $65.8 \pm 1.0\%$ and PPO = $66.1 \pm 0.4\%$ overall survival ($71.7 \pm 0.8\%$ and $71.8 \pm 1.2\%$ clean). The small across-seed standard deviation shows the headline result is not a single-seed artefact. The two agents are statistically indistinguishable on overall survival, so we do *not* claim one dominates the other (Appendix Fig. 8).

The multi-seed result is deliberately lower-budget than the headline 1M run, so it is not used to replace Table 5. Its purpose is variance calibration: if the same algorithm changed by five or ten survival points across seeds, the 1M result would be suspect. Instead, both DQN and PPO remain within a narrow band. This makes the modest expert-matching result credible even though the confidence intervals in the 1,000-episode evaluation overlap.

7 Creative Extensions

Our extension is a coherent *clinical reliability and interpretability* suite built on the existing environment, organised into four themes. The goal is not to add arbitrary complexity, but to answer a practical clinical-RL question: if a policy appears to work, can we explain *why*, can we see when it fails, and can we tell whether it is improving survival rather than merely exploiting the reward penalty?

7.1 Reward design: SOFA potential-based shaping

Under the sparse reward the agents optimise the *easy* part of the return — they lower treatment intensity (less λ penalty) rather than improve the delayed survival signal. We therefore add potential-based reward shaping [9] with potential $\Phi(s) = -\beta \text{SOFA}(s)$, rewarding transitions that reduce organ failure. Being potential-based, the shaping provably leaves the optimal policy unchanged; it is applied at *training* time only, while evaluation always uses the true reward. Shaping is what lifts the agents from near-random survival to the expert-matching results in Table 5: without it, the deep agents sit at the random survival rate and merely treat less.

SOFA is a clinically natural potential because it summarises organ dysfunction, and reducing organ failure is a plausible intermediate sign that the eventual survival outcome is becoming more likely. The shaping term therefore gives the learner a denser training signal without changing the evaluation target. In practice, we use $\beta = 0.05$ in the final run: large enough to reward meaningful SOFA improvements, but small enough that the terminal survival outcome remains the dominant objective. This design directly addresses the main weakness observed in the unshaped runs: sparse terminal rewards leave the gradient dominated by the treatment-penalty term.

7.2 Learning from demonstrations

We implement demonstration learning in both the imitation and the warm-start sense, the demonstration pre-training idea of DQfD [10], using the AI Clinician as the demonstrator [2].

Behavioural cloning (Config B). A small MLP is trained to imitate the expert’s action from the 47-dimensional observation (36k demonstration steps). On clean episodes BC matches the expert (71.7% vs 71.3% survival, intensity 0.28 vs 0.31); on the full clinical distribution it loses ~ 2 points, because it must act on the *corrupted* observation while the expert sees the true state — a clean illustration of the cost of observation degradation.

BC is included for two reasons. First, it is a sanity check that the observation vector contains enough information to recover a meaningful treatment pattern from expert behaviour. Second, it separates imitation from reinforcement learning: where BC does well, the expert policy is learnable from observations; where BC fails, the problem is not only optimisation but also distribution shift under noise/missingness. Its full-distribution survival is below the expert, which is exactly what we would expect when a supervised policy is forced to generalise to corrupted observations.

Expert warm-start (Config A). The tabular Q -table is seeded from expert demonstrations via Monte-Carlo returns, then refined by Q-Learning. The cold agent (zero initialisation) stalls near the random plateau and never reaches 70% survival in 15k episodes (68.8%, intensity 0.452); the warm-started agent escapes the plateau, reaching 70% by ~ 9 k episodes and ending at **71.9%** with a markedly more parsimonious policy (intensity 0.345), inheriting the expert’s economical treatment style (Appendix Fig. 5).

This tabular demonstration experiment is intentionally simple: it does not change the algorithm, the environment or the evaluation seeds. Only the initial Q -values change. The gain therefore isolates the value of prior clinical structure. In a sparse-reward medical task, demonstrations do not replace RL, but they provide a better starting point for exploration.

7.3 Interpretability

We explain the DQN with two complementary methods. Permutation importance measures the sensitivity of the chosen action’s Q -value to perturbing each feature; SHAP [12] attributes the greedy-action value via a gradient explainer. Both rank established sepsis-severity markers — lactate, respiratory rate, urine output — among the most decision-relevant features, which lends clinical credibility. We report permutation importance with the explicit caveat that, on a strongly correlated feature set, single-

feature permutation produces off-manifold inputs and can mis-state importance [13]; the agreement with SHAP strengthens the reading (Appendix Fig. 10).

The interpretability analysis is not presented as proof that the policy is safe. It is a diagnostic check. A policy whose decisions were driven by arbitrary feature indices or observation-scale artefacts would be suspicious even if its survival metric looked acceptable. Seeing clinically plausible markers among the largest contributors gives confidence that the network is at least using the same type of severity information that clinicians would monitor. The correlated-feature caveat remains essential: ICU variables move together, so importance should be read as a qualitative explanation rather than a causal attribution.

7.4 Q-value overestimation diagnostic

Q-Learning with a max operator on approximate values is known to overestimate [8]. Comparing the DQN's predicted value of the greedy action against the realised Monte-Carlo return over 300 episodes gives a mean predicted value of 0.830 versus a realised return of 0.636 — a positive bias of +0.194 ($\sim 30\%$). This is a direct empirical demonstration of the overestimation discussed in the value-based lecture and the standard motivation for Double DQN, listed as future work (Appendix Fig. 11).

7.5 Treatment intensity versus survival

The intensity–survival Pareto plot on the title page summarises the whole project clinically. Policy Iteration dominates (78.8% survival at intensity 0.16); the random policy is dominated by every learned policy (it treats more and survives less); and within Configuration B there is a clear trade-off — A2C is the most parsimonious (0.15), DQN the most effective on survival (0.37), with the expert, PPO and BC in between. This makes the purpose of the λ penalty explicit and frames the clinical reading of each policy.

8 Discussion

Why the random baseline is high. *ICU-Sepsis* has, by design, a high random-policy survival rate and a small optimal-versus-random gap: the cohort is drawn from real trajectories where most patients survive, so the action has a small effect relative to the patient's initial state. Our own Policy Iteration confirms the ceiling — 78.8%, only ~ 10 points above random. Beating random by a large margin is therefore structurally impossible, and any claim of, say, 90% survival would indicate a changed environment rather than a better agent. The value of the project is in the *analysis*, not in an inflated number.

Parsimony versus survival. The Pareto and shaping results together give the sharpest insight: naive deep RL on the sparse reward learns to *treat less* (raising return through the penalty) rather than to *save more patients*. Reward shaping redirects the gradient toward survival and is what makes the agents match the expert. We state this transparently rather than presenting the shaping-dependent numbers as if they arose from the bare objective.

Significance. The deep agents, the expert and each other are largely within overlapping confidence intervals on overall survival; the clearer separations are on clean episodes ($\approx 74\%$ vs $\approx 69\%$ random) and in the across-seed study, where the small standard deviations ($\leq 1\%$) make the modest expert-matching gains credible rather than noise.

Connection to theory. The project instantiates the full arc of the course: dynamic programming (Policy Iteration), on- and off-policy temporal-difference control (SARSA, Q-Learning), value-function approximation and DQN, policy-gradient and actor-critic methods (PPO, A2C), the overestimation problem (the +0.194 diagnostic), value-function approximation versus tabular representation (the very reason Configuration B needs deep RL), and reward design (potential-based shaping). Methodologically

it follows the guidance for clinical RL [4] — a clinician baseline, uncertainty quantification, multiple seeds [11], and in-simulator evaluation that avoids off-policy estimation on logged data.

Configuration A versus Configuration B. The cross-configuration comparison should not be read as “deep RL is worse than tabular RL” in general. It shows what changes when the problem becomes more realistic. Configuration A exposes the full MDP and gives an integer state, so the best method is exact planning. Configuration B hides the model, replaces the state id with a high-dimensional clinical vector, and injects observation failures. The lower survival is therefore not surprising; it is the cost of moving from a clean known-MDP classroom setting to a deployment-style setting.

Clinical reading. The best learned policies are useful because they are not uniformly more aggressive. DQN improves survival with moderate intensity; A2C achieves high return mostly through very low intensity; Policy Iteration shows that the optimal tabular policy can be both effective and parsimonious when the model is known. This is the right clinical framing: an RL policy for sepsis is not valuable merely because it recommends treatment, but because it balances outcome and treatment burden under uncertainty.

9 Limitations and Future Work

The hyperparameters were tuned with a short Optuna proxy (15 trials, 50k steps), which is noisy; the perceptual benefit of longer training is not exhausted. The multi-seed study uses 300k steps for tractability rather than the full 1M. The expert acts on the true state while the agents act on (possibly corrupted) observations, so the expert comparison is a reference rather than a strict upper bound. Three concrete extensions follow: (i) a **Double DQN** comparison to test whether decoupling action selection from evaluation reduces the measured $+0.194$ overestimation and improves survival; (ii) a **distributional** agent (QR-DQN) as a fourth deep method; (iii) a β sweep of the SOFA shaping to trace its sensitivity. Finally, all results are on a *simulated* benchmark and are **not** clinical advice.

Two additional limitations are worth stating explicitly. First, the simulator is derived from observational data, so its transitions inherit modelling assumptions from the ICU-Sepsis/AI Clinician pipeline; policy performance in the simulator is not evidence of bedside efficacy. Second, the failure-mode wrappers are useful stress tests but simplified versions of real data quality problems. Real ICU missingness is often informative, correlated with workflow and patient severity, and not necessarily well represented by zeroing a fixed subset of features. These limitations do not invalidate the benchmark result, but they define the boundary between a course project and a deployable clinical system.

10 Conclusion

On *ICU-Sepsis*, model-based Policy Iteration is the clear Configuration A winner (78.8% survival at the lowest intensity), with tabular Q-Learning and SARSA beating random but trailing the exact optimum, exactly as DP-versus-TD theory predicts. In the harder continuous Configuration B, DQN, PPO and A2C — with observation normalisation, SOFA reward shaping and tuned hyperparameters — match the AI Clinician on overall survival and exceed it on clean episodes, are stable across seeds, and exhibit a measurable value-overestimation bias. A reward-design, learning-from-demonstrations and interpretability suite, anchored by an intensity–survival Pareto analysis, shows that the central challenge is the sparse reward driving parsimony over survival, and that shaping resolves it. The project covers the theoretical and practical content of the course end to end while respecting the methodological standards of the clinical-RL literature.

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A Appendix: Compact Figure Reference

All figures are reproducible from the committed result files and the `run_*.py` scripts in the repository. They are shown compactly here so the page budget is spent mainly on methodology and interpretation.

A.1 Configuration A

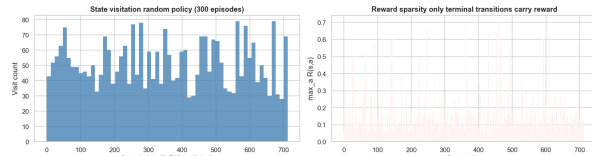


Figure 1: Config A environment exploration.

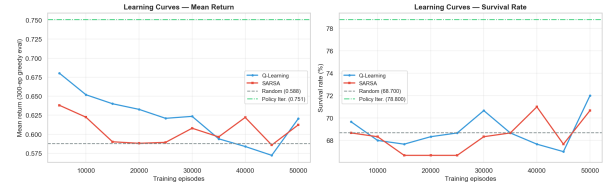


Figure 2: Config A learning curves.

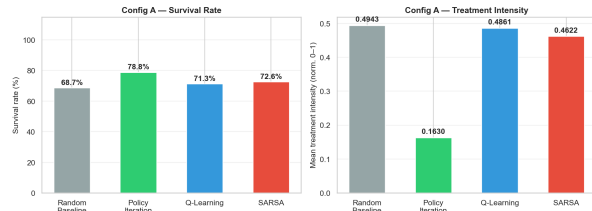


Figure 3: Config A survival and intensity.

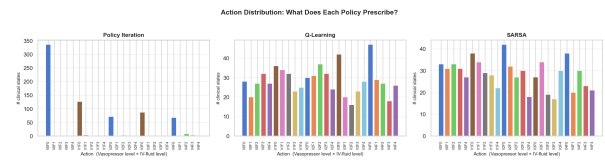


Figure 4: Config A action distributions.

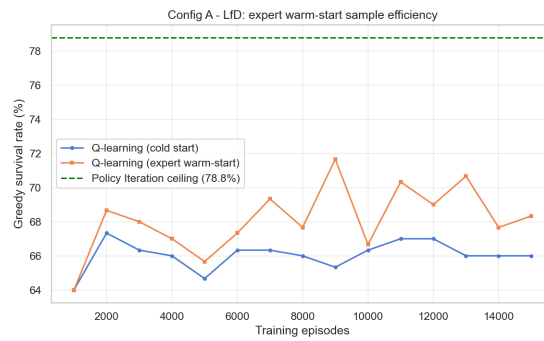


Figure 5: Expert warm-start versus cold Q-Learning.

A.2 Configuration B



Figure 6: Config B 1M learning curves.

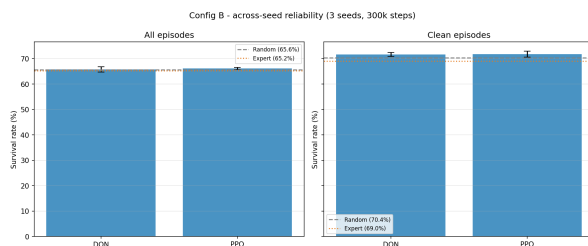


Figure 8: Across-seed reliability.

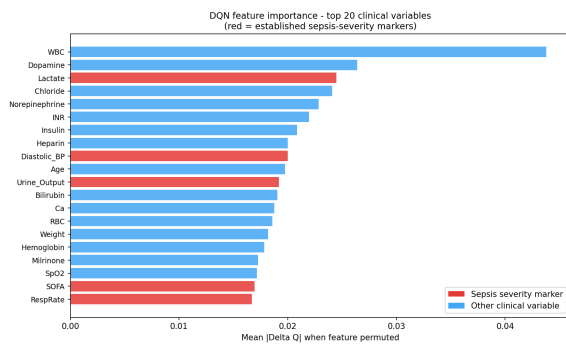


Figure 10: DQN feature importance.

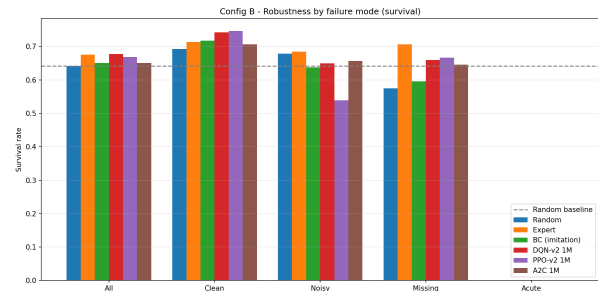


Figure 7: Config B survival by failure mode.

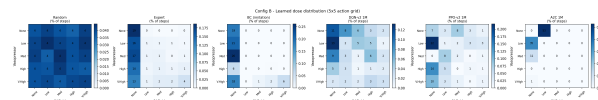


Figure 9: Learned vasopressor/fluid dose grids.

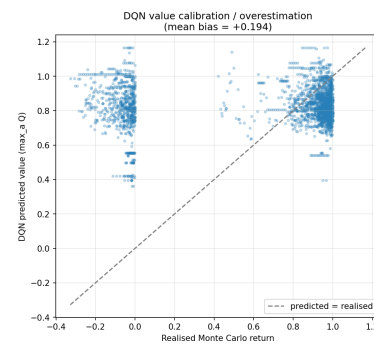


Figure 11: DQN value overestimation diagnostic.